

Theory Concepts

Real Estate House Price Prediction — Supervised Learning & Regression Project

Part A: Conceptual Understanding

1. What are Supervised Learning Algorithms?

Supervised learning algorithms learn a mapping from input features (X) to a known output/label (y) using a labeled training dataset. The model is trained by comparing its predictions to the true labels and adjusting its internal parameters to minimize the error. Once trained, it can predict outputs for new, unseen inputs. Common examples include Linear Regression, Logistic Regression, Decision Trees, and Support Vector Machines.

2. Difference between Regression Algorithms vs Classification Algorithms

Regression algorithms predict a continuous numeric value (e.g., house price, temperature) and are evaluated using metrics like MSE, MAE, and R^2 . Classification algorithms predict a discrete category or class label (e.g., spam vs. not-spam) and are evaluated using metrics like accuracy, precision, recall, and F1-score. In this project, house price prediction is a regression task since price is a continuous quantity.

3. Explain Simple Linear Regression

Simple Linear Regression models the relationship between a single independent variable (X) and a dependent variable (y) using a straight line: $y = \beta_0 + \beta_1 \cdot X + \epsilon$, where β_0 is the intercept (value of y when $X = 0$), β_1 is the slope (change in y per unit change in X), and ϵ is the error term. The best-fit line is found by minimizing the Sum of Squared Errors (SSE) between predicted and actual values — typically via the Ordinary Least Squares (OLS) closed-form method or iteratively via Gradient Descent.

4. List and explain the Assumptions of Linear Regression

- **Linearity** — the relationship between the independent and dependent variables is linear.
- **Independence of errors** — residuals are not correlated with one another (no autocorrelation).
- **Homoscedasticity** — residuals have constant variance across all values of the independent variable(s) (no funnel/cone pattern).
- **Normality of residuals** — residuals are approximately normally distributed.
- **No/low multicollinearity** — in multiple regression, independent variables should not be highly correlated with each other.
- **No significant outliers or high-leverage points** that unduly influence the fitted line.

5. What is Bias–Variance Trade-Off?

Bias is the error introduced by approximating a complex real-world relationship with an overly simple model; high bias leads to underfitting, where the model performs poorly on both training and test data.

Variance is the error introduced by a model's sensitivity to small fluctuations in the training data; high variance leads to overfitting, where the model performs very well on training data but poorly on unseen data. As model complexity increases, bias tends to decrease while variance tends to increase. The bias–variance trade-off is the search for the complexity level that minimizes total error ($\text{bias}^2 + \text{variance} + \text{irreducible noise}$), i.e., a model that generalizes well to new data.

6. Explain Overfitting and Underfitting with examples

Underfitting occurs when a model is too simple to capture the underlying pattern in the data — for example, fitting a straight line to data with a clearly curved relationship. Both training and test error remain high. Overfitting occurs when a model is too complex and starts memorizing noise in the training data instead of the general pattern — for example, fitting a very high-degree polynomial (e.g., degree 8) to house price data. Training error becomes very low, but test error rises because the model fails to generalize to new houses it has not seen before.

Full practical implementation, code, evaluation results, and figures for all remaining parts (B–I: Dataset Preparation, Simple & Multiple Linear Regression, Polynomial Regression, Gradient Descent Optimization, Bias–Variance Diagnostics, and Final Reporting) are available in **House_Price_Prediction_Project.ipynb** in this repository.